

Andrew Park

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EDUCATION

University of Illinois at Urbana-Champaign, Grainger College of Engineering

Bachelor of Science in Mechanical Engineering; Minor in Spanish

Expected May 2027

Cumulative GPA: 3.96/4.0

Relevant Coursework: Robotics, Design for Manufacturability, Vibrations, Materials, Compressible Flow, Aerodynamics

EXPERIENCE

Google, Product Design Engineering Intern – **Chicago, IL**

May 2026 – August 2026

- Led the development, design, and validation of a novel EMI shielding method from concept through prototype evaluation
- Fabricated experimental setups and worked with cross-functional teams to conduct tests to characterize various adhesives
- Created multiple shield can designs in Siemens NX, collaborating with suppliers for DFM for sheet metal drawing
- Developed and executed a drop-in shield prototype campaign for the validation of new shielding technologies, designing and sourcing multiple custom components and carrying out manual retrofitting operations on baselined legacy devices
- Constructed a modifiable assembly sequence for a main logic board in NX, accommodating unique and complex geometries, facilitating manufacturing workflows and defining design keep outs

SpaceX, Launch Engineering Intern – **Vandenberg, CA**

Feb 2026 – April 2026

- Owned end-to-end design, build, and activation of the first pneumatic pressure testing facility at the Vandenberg launch site
- Facilitated location transition of ground support equipment (GSE) proofing and assembly by providing pressure testing capabilities up to 10,000psi, saving \$3500/month in shipping and 570h/month in labor
- Planned and executed a first-of-a-kind on-site pressure test of a removed flight component, coordinating LOX cleanliness controls, blast energy mitigations, and acoustic clears to successfully verify the component for return to service
- Designed, validated, and released P&ID diagrams for high-pressure pneumatic GSE panels handling GHe and GN2 utilizing an air-driven boost pump system, incorporating preliminary and critical design reviews with senior engineers
- Oversaw physical panel build, facility layout, and GSE installation, including coordination with contractors for wall removal and replacement for installation of a 4,000lb boom shed on a 1-week timeline
- Developed and executed acoustic testing and analysis on boom shed setup, coordinating with engineering health and safety (EHS) teams to establish safety protocols including blast energy limits and physical clear distances

Caterpillar, Inc., Design Engineering Intern – **Pontiac, IL**

May 2025 – August 2025

- Spearheaded the design and development of a real-time temperature monitoring system for solenoids, enabling early detection of thermal failure and significantly reducing unplanned downtime and costly replacements
- Engineered a custom microcontroller-based solution with 1°C accuracy and 32ms sampling, interfacing with relays, current sensors, a DAQ, and private display software using USB and other communication methods
- Designed and assembled a new fixture adapted to unique injector geometries in CREO, facilitating disassembly by allowing users to bypass having to release torque on key components

Argonne National Laboratory, Science Undergraduate Laboratory Intern (SULI) – **Lemont, IL**

May 2024 – August 2024

- Increased precision by 82% and run time by 60% on a post-processing python script detecting undesirable behavior in sustainable fuel combustion testing through analysis of pressure transducer data
- Identified the start of compression on Rapid Compression Machine experiments within 0.1ms by fitting experimental pressure curves to models of isentropic compression based on Linear Variable Differential Transformer (LVDT) data
- Compared results from pressure and LVDT datasets to evaluate how to account for noise and piston seating time while detecting asynchronous behavior, improving accuracy by 90%

PROJECTS

6-Axis Robot Arm, *American Society of Mechanical Engineers (ASME)*

October 2024 – April 2025

- Led a team of over 40 students to design a 2-foot, 6-DOF robotic arm loosely inspired by the Modern Robotics UR3 for presentation at the University of Illinois's Engineering Open House (EOH) event on a budget of \$300
- Transmitted four distinct PWM signals to simultaneously control 4 stepper motors by manually calculating the output compare register value with different timer frequencies and prescales on clear timer on compare match (CTC) mode
- Implemented a closed-loop PID control system for 2 DC motors to run in tandem with PWM stepper control
- Generated forward and inverse kinematics in Python to generate sets of motor angles from 3D positions and orientations

SKILLS

Design: CREO, Siemens NX, Fusion 360, P&ID, Microcontrollers, DFM, DFA

Analysis & Control: C/C++, python, ROS, MATLAB, aPriori, Java, Simulink, Ansys Fluent